Human Movement Science 7 (1988) 27–46
North-Holland

SELF-ORGANIZATION OF COORDINATIVE MOVEMENT PATTERNS *

J.A.S. KELSO and G. SCHÖNER
Florida Atlantic University, Boca Raton, USA

Kelso, J.A.S. and G. Schöner, 1988. Self-organization of coordinative movement patterns. Human Movement Science 7, 27–46.

We present a strictly operational approach – in which theoretical tools and experimental data are developed together – to the problem of understanding the coordination of movement patterns. The empirical aspects are guided by synergetics, a theory of spontaneous pattern formation in open systems. Following an outline of our theoretical strategy, recent experimental results are reviewed that demonstrate the validity of the approach. From these studies, it is possible to establish the linkage between order–order transitions in movement behavior and other nonequilibrium phase transitions in nature. Behavioral patterns (corresponding to low-dimensional attractors for collective variables) and their dynamics are shown to arise in a purely self-organized fashion from cooperative coupling among individual components. This step has been implemented analytically and computationally. The insights gained from the present research allow a generalization in the form of seven theoretical propositions that aim at characterizing pattern formation, stability and change in complex, biological systems. In turn, a number of new research directions emerge, including studies of the collective dynamics of the environment–movement system, learning, and multiple-limb coordination.

1. Brief prologue: The synergy concept

The concept of synergy has been a powerful one in the field of behavioral physiology. In the context of neuromuscular coordination it

* An earlier version of this paper was prepared for the Soviet-American Symposium on Motor Control, Moscow, March 22–April 1, 1987 which unfortunately we were not able to attend. A later version was given by JASK as an invited address to the North American Society for the Psychology of Sport and Physical Activity, Vancouver, B.C., Canada, June 4–7, 1987.

The present work was supported by a Joint Contract of the U.S. Office of Naval Research and the U.S. Air Force Office of Scientific Research and National Institutes of Mental Heath Grant 1PO1MH42900-01. G. Schöner was supported by a Forschungsstipendium of the Deutsche Forschungsgemeinschaft, Bonn, FRG.

Requests for reprints should be sent to J.A.S. Kelso, Center for Complex Systems, Building Mt9, Florida Atlantic University, Boca Raton, FL 33431. USA.

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appears to have evolved in the late 18th or early 19th century from its more general usage in medical science as the combined or correlated action of two or more drugs in the treatment of disease. From stimulation studies of ventral nerve roots in monkeys, the English physiologists, Ferrier and Yeo concluded that elicited movements were 'not mere contractions more or less strong of the various muscles, but a highly coordinated functional synergy in each case' (1881 : 15). Though these experimental results were refuted later by Sherrington (1892), the synergy concept – as a complex combination of muscle actions which occurred for a definite purpose – remained. For Sherrington, however, the source of synergies lay in spinal and brainstem *reflexes* which constituted the basis of all action: 'With the nervous system intact the reactions of various parts of that system, the "simple reflexes" are ever combined into great unitary harmonies, actions which in their sequence one upon another constitute in their continuity what might be termed the "behavior" of the individual as a whole' (Sherrington 1906 : 239). How 'the motor paths at any moment accord in a united pattern for harmonious synergy, cooperating for one effect' (Sherrington 1906) remained a mystery. In more modern times, our Russian colleagues, following Bernstein and led by Professor Gelfand offered a linguistic metaphor for synergies and a powerful visual image. For them, synergies 'form a dictionary of movements in which the efforts of the muscles are the letters of the language, the synergies combine these letters into words, the number of which are much less than the number of combinations of letters' (Gelfand et al. 1971 : 332). 'The language of synergies' for Gelfand et al., is 'not only the external language of movements, but also the internal language of the nervous system during the control of movement' (p. 333).

Synergies thus are patterned, cooperative states arising in a neural medium. But how are they formed? How is a synergy assembled from its components? And what principles govern synergy assemblage? We start out here, as we did some years ago (e.g., Kelso et al. 1980; Kugler et al. 1980) with a rather vague notion, that a synergy (an ensemble of neuromuscular components temporarily assembled as a task-specific unit) is a dissipative or synergetic structure (a pattern established away from equilibrium). Then we present a specific case (a nonequilibrium phase transition in movement) where the idea can be given a concrete and precise meaning. From that case, we gradually generalize until we end up with an idea no less universal than the one from which we

began. Only now the idea is no longer vague, but mathematically exact. The identity between a synergy or coordinative structure (more generally, the self-organized formation of coordinated movement patterns) and a synergetic structure can therefore be established.

2. Introduction: From synergies to synergetics

The movements of animals, including people are ordered spatio-temporal structures that arise in a system composed of very many neural, muscular and metabolic components that operate on different intrinsic time scales. The order is such that we are often able to classify it, like the gaits of a horse, for example, or the limited set of basic sounds (the so-called phonemes) that are language-universal.

Structure or pattern somehow emerges from complexity in a fashion reminiscent of the spontaneous formation of structure in open, non-equilibrium systems. In such physical, chemical and biochemical systems, it has been shown more and more that ordered patterns are produced in a self organized fashion. Indeed there is growing evidence that even though the material substance that can realize such patterns is quite variable, the underlying principles may be the same (see Haken (1983) for review). It is partly because of its foundations in physics and partly because certain recently discovered ordering phenomena in movement are open to explicit nonequilibrium systems analysis, that we were led to adopt such a theoretical approach.

In the following we first define our theoretical strategy to biological coordination and illustrate it through studies of movement patterns. Although many of the details are published elsewhere, we emphasize new results that lead, we believe, to a coherent understanding of movement coordination, especially with respect to stability and change of dynamic pattern. Then we summarize the insights derived from such studies in a set of seven theoretical propositions that aim at a more general characterization of pattern stability and change in biological systems. The present approach appears to afford extension to: different movement systems (e.g., speech; Kelso et al. 1986a); different levels of description (e.g., neuronal; Kelso et al. 1987); the collective dynamics of perception–action patterns and learned movement patterns (Schöner and Kelso in press-a,b) and to developing as well as mature, skilled behavior (Thelen et al. 1987). In addition, certain new biologically

relevant problems are motivated by insights gained into the proposed organizational principles, some of which we raise at the end of the paper.

Why should an understanding of complex biological systems and the behavioral complexity that they exhibit be sought in terms of organizational principles rather than mechanisms? One reason is that biological systems generate ordered behavior in a *flexible* (e.g., changing according to environmental demands) as well as *stable* fashion (e.g., persisting for a certain amount of time under varying environmental conditions). There is mounting evidence that such features are characteristic of softly assembled cooperative states, not hard-wired mechanisms (e.g., Kelso et al. 1984). An other reason is that well-defined behavioral functions (e.g., locomotion) can be supported by many different material structures, yet at the same time a given material structure can support many different behavioral functions. Speaking and chewing, for instance, use overlapping sets of articulators to accomplish different functional requirements.

A unique feature of our approach is to characterize the dynamics of the patterns themselves at different observational levels and in different systems (see below). The theoretical backdrop comes from *synergetics* (e.g., Haken 1983), originally a physical theory of spontaneous formation of structure in open systems (e.g., in lasers, hydrodynamics, chemical instabilities), but which now encompasses other fields as well. In open systems structure or change of structure occurs spontaneously (i.e., in a *self-organized* fashion) at so-called nonequilibrium phase transitions, when environmental conditions as expressed by control parameters reach a critical point. It has been shown that in such situations these systems are governed by only very few degrees of freedom (this is also called the *slaving principle*, cf. Haken 1983). Synergetics thus promotes a search for the essential, collective variables of the system under study (the so-called order parameters) whose nonlinear dynamics can be defined in terms of low-dimensional *attractors*. The existence of an attractor means on the one hand that there has been a tremendous reduction of complexity (viewed from the number of degrees of freedom involved). On the other hand, a significant feature of low-dimensional attractors is that they are capable of generating a considerable degree of *behavioral complexity*. That is, very complicated phase diagrams can occur including transitions to deterministic chaos (see e.g. Lundquist 1985). Thus, some of the very

features that one wants to explain – the compression of degrees of freedom, the capability of a system to exhibit multiple behavioral patterns, and to switch among them – can be understood using attractor dynamics.

However, a general across the board application of synergetics to coordinated movement behavior is hazardous. Rather, an operational approach – linking experimental observations to theory – is to be preferred (Haken 1985). In contrast to certain physical systems like the laser, for instance, in biology the path from the microscopic dynamics (e.g. the brain with 10^{14} neurons and neuronal connections) to collective order parameters for macroscopic behavior is not readily accessible to theoretical analysis. On the other hand, using what we call a 'synergetic strategy' (Kelso and Schöner 1987; Kelso et al. 1987) it may still be possible to understand biological coordination. The main features of this strategy, which, we emphasize again, requires an intimate link between experiment and theory, are:

(1) the identification of collective variables (order parameters) that characterize movement patterns;
(2) the mapping of observed stable patterns onto attractors of the collective variable;
(3) The determination of 'control' parameters that move the system through its collective states;
(4) the study of stability and loss of stability of movement patterns, which allow a determination of the collective variable's dynamics;
(5) the establishment of the relationship among levels of observation by deriving collective variable dynamics from cooperative coupling among components.

Using this strategy, which can be implemented most successfully near points of pattern change, i.e., where differential characterization of pattern is possible, a number of existing results on coordination of multidegree of freedom movements can be explained, new phenomena predicted and subjected to experimental test. Some of these we present next.

3. From phenomena to models to predictions to understanding

A good experimental test field for the synergetic strategy in biology is multidegree of freedom movements. These are rich enough in their

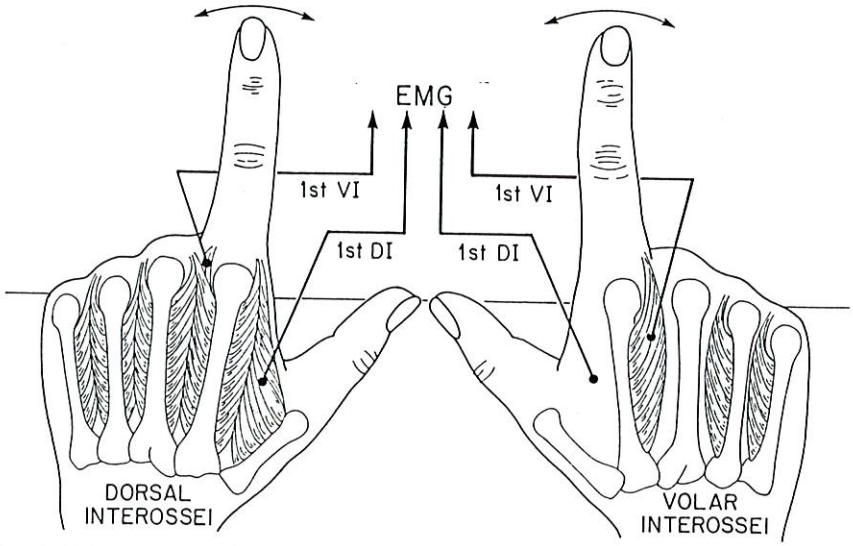


Fig. 1. Subjects move their index fingers rhythmically in the transverse plane with the same frequency F for left and right finger. The movement is monitored by measuring continuously the position of LEDs attached to the finger tips with a modified SelSpot-camera system. The electromyographic (EMG) activity of the right and left first dorsal interosseous (FDI) and first volar interosseous (FVI) muscles are obtained with platinum fine-wire electrodes (drawing by C. Carello).

dynamics, yet sufficiently accessible to observation – in part because of recent technological developments – that they are open to analysis. Indeed, as mentioned earlier, it was the discovery of a phenomenon – a nonequilibrium phase transition in biological motion – that led us to adopt the present approach in the first place. Let's briefly describe the original finding (Kelso 1981 1984).

The task for human subjects is to move their index fingers or hands rhythmically (see fig. 1). The driving frequency, F (in Hz), corresponding to the control parameter is then increased in a stepwise fashion. In such a situation subjects can stably and reproducibly perform in only two phase-locked modes, either in-phase (homologous muscle groups contracting simultaneously) or anti-phase (homologous muscle groups contracting in an alternating fashion). The finding that only two stable phase-lockings exist was demonstrated quantitatively in experiments in which the relative phase between the two fingers was manipulated and the variability of the produced relative phase was measured. Much less

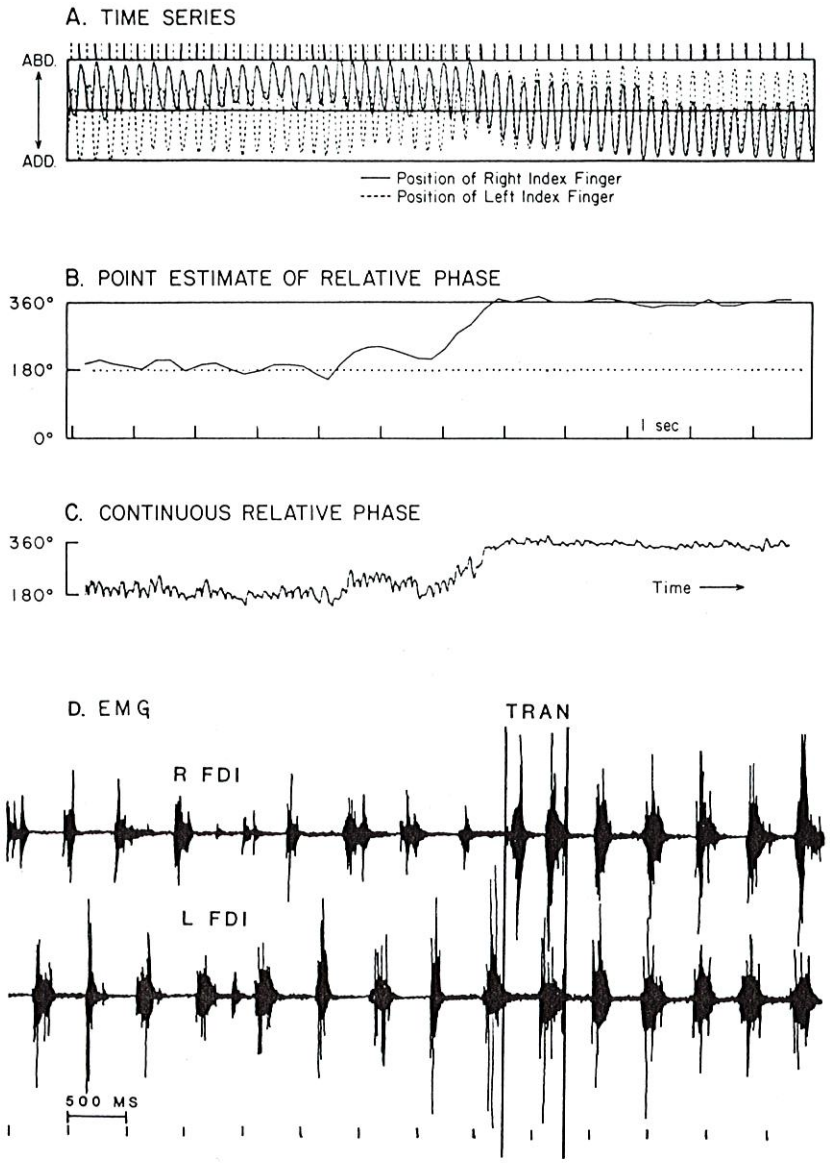


Fig. 2. A: The time series of left and right finger position shows the transition from anti-phase movement (left portion) to in-phase movement (right portion). From left to right the movement frequency, F was increased. B: The point estimate of relative phase (obtained from the relative position of the left finger's peak extension in the right finger's cycle) changes from fluctuating around 180 deg to fluctuating around 360 deg. C: A more refined measure of relative phase is the continuous estimate (obtained from the difference of the individual finger's phases, that were calculated from the phase plane $(x, \dot{x})$ trajectory). D: The EMG record of left and right FDI also show the change in phasing.

variable performance was observed at relative phase $\phi = 0$ rad (in-phase) and $\phi = \pm\pi$ rad (anti-phase) than any other specified relative phase (Tuller and Kelso 1985; Yamanishi et al. 1980). In the Kelso experiments, when subjects, initially moving in the anti-phase mode, were instructed to increase cycling frequency, an involuntary abrupt shift to the in-phase mode was observed at a critical frequency. Beyond this critical frequency value only the in-phase mode was performed stably (see fig. 2). This result has been reproduced by several other groups of investigators (e.g., MacKenzie and Patla 1983; Baldissera et al. 1982) using different experimental paradigms and various anatomical components. Earlier Cohen (1971) observed transient behavior between the two coordinative modes, but did not manipulate the control parameter (frequency), and hence did not establish the transition.

It seemed at least intuitive that the foregoing experimental observations were consistent with other bifurcation phenomena in nature (Kelso 1984). Indeed, their essential features were successfully modelled by Haken et al. (1985), introducing relaxational dynamics

$$\dot{\phi} = \frac{dV(\phi)}{d\phi}, \tag{1}$$

for the relative phase ϕ and mapping the observed modes onto point attractors. The potential

$$V(\phi) = -a\cos(\phi) - b\cos(2\phi), \tag{2}$$

complying with periodicity and symmetry requirements, captures the bifurcation diagram in that it has minima at $\phi = 0$ and $\phi = \pm\pi$ rad for $b/a < 0.25$ with the latter minimum turning into a maximum for $b/a > 0.25$ (see fig. 3). All the main features of the experiments – the presence of only two stable relative phase or attractor states between the hands; the transition from one attractor state to the other at a critical cycling frequency; the existence of only one attractor state beyond the transition; hysteresis, i.e., when cycling frequency is reduced the system stays in the in-phase mode – are reproduced by this minimal model.

How can the individual components give rise to phase-locked temporal patterns? Haken et al. (1985) have determined the simplest coupling structure that can account for the observed phase-lockings. It

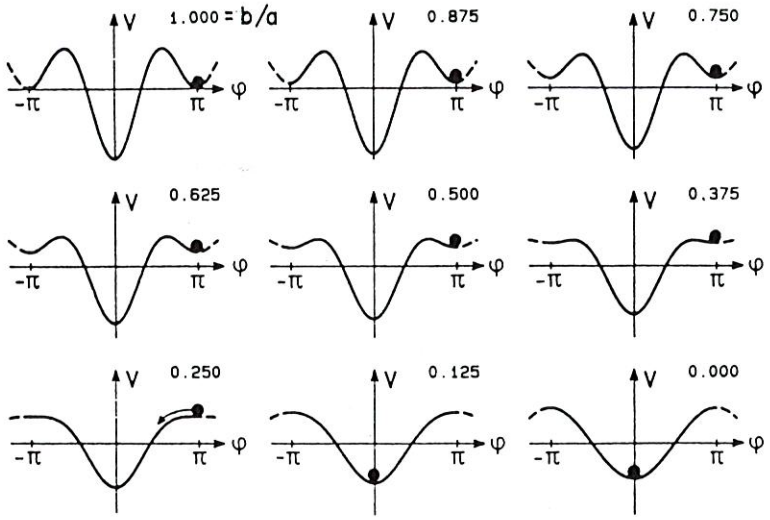


Fig. 3. The potential (2) as the ratio b/a is changed. The little ball illustrates the behavior of the system if it is initially (upper left corner) prepared in the anti-phase state.

is worth emphasizing that precisely the same theoretical strategy was applied at the single component level (i.e., to the study of *individual* rhythmic hand movements) as at the level of the collective variable, relative phase. That is, it was again possible to map certain experimentally observed features of individual rhythmic movement patterns onto limit cycle attractors. Among the former were: (a) reproducible kinematic relations between amplitude, frequency and velocity of movement (Kay et al. 1987); (b) stability, as observed in perturbation experiments (Kay 1986; Kelso et al. 1981); and (c) autonomy, as seen in detailed phase resetting experiments (Kay 1986). The functional form of the limit cycle attractor, dictated by the foregoing constraints, is a combination of the well-known Van der Pol and Rayleigh oscillators:

$$\ddot{x} + f(x, \dot{x}) = 0, \quad \text{with} \tag{3}$$

$$f(x, \dot{x}) = \alpha\dot{x} + \omega^2 x + \beta\dot{x}^3 + \gamma\dot{x}x^2, \tag{4}$$

where $\alpha < 0$, $\omega > 0$, $\beta > 0$ and $\gamma > 0$ are model parameters.

The most simple coupling structure that accounts for both phase-locked coordinative patterns as well as the transition from one to the other at a critical frequency is:

$$\ddot{x}_1 + f(x_1, \dot{x}_1) = (\dot{x}_1 - \dot{x}_2)\left[a + b(x_1 - x_2)^2\right], \tag{5}$$

$$\ddot{x}_2 + f(x_2, \dot{x}_2) = (\dot{x}_2 - \dot{x}_1)\left[a + b(x_2 - x_1)^2\right], \tag{6}$$

where a and b are coupling constants (cf. Haken et al. 1985; Kay et al. 1987). In the limit of weak nonlinearity, and for different coupling strengths, the relative phase dynamics (1), (2) were derived by Haken et al. (1985; see also Kelso and Scholz (1985) for computer simulations). Again, experimental observations showed that kinematic relations (e.g., between amplitude and frequency) were not significantly different between the coordinative modes and the single hand movements, indicating that the coupling constants a, b were small compared to the corresponding coefficients α, γ of the oscillator function (4) (see Kay et al. 1987). Yet phase-locking still occurred. Our main point here, however, is not the mathematical aspects, per se, but the *minimality* strategy adopted at both the level of the collective variable dynamics and the level of the individual component dynamics. Both are low-dimensional descriptions – in terms of dissipative dynamics – that account for the observed facts in a unified manner. In this fashion (a) a rigorous relation between levels of observation, and (b) a corresponding compression of degrees of freedom, may be established.

The foregoing modelling work inspired further attempts to explicitly test the phase transition character of the phenomenon. For example, a stochastic generalization of the model defined in eqs. (1), (2) by Schöner et al. (1986), led to additional predictions regarding certain characteristic features, including *critical fluctuations* of the hypothesized order parameter and *critical slowing down.* Although these predictions are theoretically dependent, they are measured independently, as we shall see. The predictions reflect the centrality of the concept of *stability* in the understanding of biological coordination not just as a characterization of the two attractor states, but also because *loss of stability* is hypothesized to be the generic mechanism effecting coordinative change.

In the Schöner et al. (1986) model, stochastic forces act as continuously applied perturbations therefore producing deviations away from

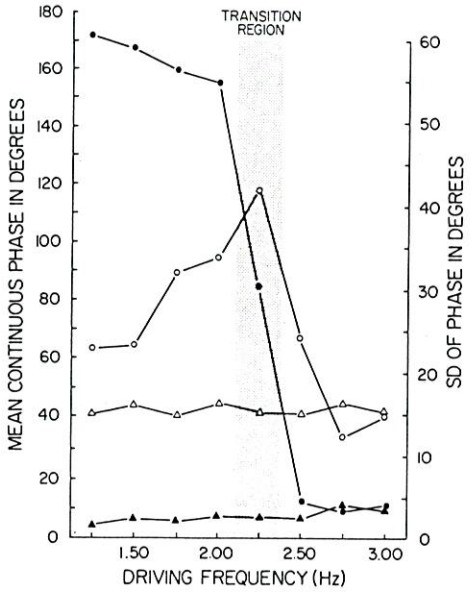


Fig. 4. The average mean relative phase modulus for the in-phase (closed triangle) and anti-phase (closed circles) modes of coordination and the average *SD* (in-phase = open triangles, anti-phase = open circles) as a function of driving frequency *F* (in Hz) for a set of 10 experimental runs. On a given run, the mean and *SD* were calculated for the last 3 sec (600 samples) at a given frequency (from Kelso et al. 1986b).

the attractor state. The size of these fluctuations as measured, for instance, by the *SD* of ϕ is thus a metric for the state's stability. A recent paper (Kelso et al. 1986b) examined the mean relative phase and its *SD* as frequency was scaled systematically in each of the two coordinative modes. When parameter scaling began in the in-phase mode, both variables remained roughly constant (see fig. 4). In contrast, a clear enhancement of fluctuations was observed for the anti-phase mode both before and during the transition. Thus the first major prediction of the stochastic dynamic model – the presence of critical fluctuations – received striking experimental confirmation.

The second main prediction from stochastic dynamic theory concerns critical slowing down. If a small perturbation is applied to the system driving it away from its stationary state, the time it takes for the system to return to that state – the so-called 'local relaxation time', τ_{rel} – is a measure of the stability of the attractor. The smaller τ_{rel}, the more stable the attractor. In the present context, as the system loses its

stability, the time it takes to reach the stationary anti-phase state following a slight perturbation should increase as the frequency of oscillation is scaled toward the transition. No strong increase in τ_{rel} is predicted for the in-phase coordinative state.

A test of this prediction (Scholz et al. 1987) required considerable modification from the experimental set-up used to study critical fluctuations. In particular, special allowance was made to incorporate the injection of perturbations. The subjects' task was again to oscillate both fingers rhythmically at the same frequency in an in-phase or, in other runs, an anti-phase mode of coordination. The movement was paced by an auditory metronome pulse with the instruction to perform one complete movement cycle per pulse. The initial pacing frequency was determined individually for each subject based on a series of 10 scaling test trials that determined the frequency of movement at which the subject switched spontaneously from the anti-phase to the in-phase mode of coordination with highest incidence. The initial pacing frequency was then chosen to be five 0.2-Hz steps below that most frequently occurring transition frequency. In the actual experimental runs the pacing frequency was increased every 10 sec in nine 0.2-Hz steps.

In order to estimate the system's relaxation time, the pattern of bimanual coordination was disturbed by perturbing the right index finger with a 50 msec torque pulse that was generated by a dc torque motor mounted above the finger's axis of rotation. The torque pulse was timed to the peak velocity of finger movement using a servo-control circuit, and occurred randomly during a frequency plateau in the flexion direction. The size of the torque pulse was individually adjusted to produce approximately comparable displacements of the right index finger. In each trial, movement was perturbed in two non-adjacent frequency plateaus, which were randomly distributed over a block of trials, such that each of the nine frequency plateaus was perturbed a total of ten times.

Using interactive computer displays, an estimate of the relaxation time was obtained from the time from the torque pulse offset until the relative phase time series stabilized at its pre-perturbation mean value. The following features were noted in all five subjects tested: (1) Except for the lowest frequencies, the relaxation time in the anti-phase mode was consistently higher than in the in-phase mode. (2) As the frequency approached the transition frequency, the relaxation time in the anti-

phase mode increased while it remained constant or decreased in the in-phase mode. A mode by pacing frequency analysis of variance performed individually for each subject's data showed that this difference was statistically significant in all but one case. Even for the single outlier, who showed an overall decrease of relaxation time in both modes, a sharp increase occurred in the anti-phase mode immediately prior to the transition. Indeed, this subject showed a very strong tendency to shift into the in-phase coordinative pattern when perturbed near the transition (see Scholz (1986). In sum, overall pretransitional increases in relaxation time prove the presence of critical slowing down in this biological coordination problem entirely consistent with stochastic dynamic theory.

Finally, stochastic dynamic theory contains a further, quite novel feature that has not been studied thus far, the so-called *switching time*, i.e., the duration of the transient from the anti-phase to the in-phase state. The basic idea is that during the transition, the probability density of ϕ – initially concentrated at $\phi = \pi$ rad – flows to $\phi = 0$ rad and accumulates there until the 'new' peak at $\phi = 0$ rad becomes dominant and stationary. In Schöner et al. (1986) predictions for the distribution and the mean duration of this process are provided that have again been recently evaluated (Scholz et al. 1987).

Interactive computer displays were used to measure the switching time on frequency plateaus in which a transition occurred. Here the estimate was determined as the time from the beginning of the frequency plateau to the point where the relative phase time series stabilized at a zero (or 2π rad) mean value corresponding to the completion of the transition.

The histogram shown in fig. 5 presents the experimentally determined switching times for the present experiment. A quite astounding agreement with the theoretical curve (also shown) is observed, both in terms of the mean switching time and the shape of the distribution. Hence yet another feature of the dynamics of biological coordination is shown to follow in detail simple stochastic dynamics for the collective variable ϕ. In particular, we find that noise is qualitatively and quantitatively relevant for the switching behavior studied here, closely paralleling similar features in physical switching processes (see Landauer (1979) for review).

To summarize, data and theory converge on a nonequilibrium phase transition interpretation (in the sense of synergetics) of the present

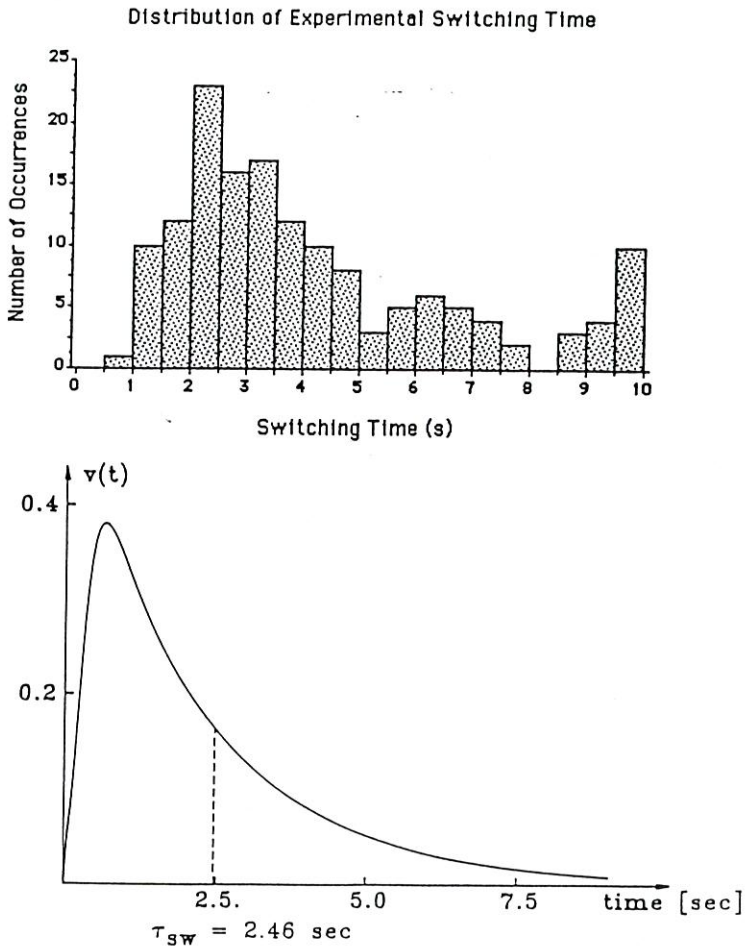


Fig. 5. Top: The distribution of switching times (time from last change of control parameter frequency to the completion of switching) from all subjects and trials. Again we discarded trials in which the transition was induced by one of the torque pulse perturbations (from Scholz et al. 1987). bottom: The distribution of switching times as calculated theoretically from the transient probability distribution of relative phase at the critical point (from Schöner et al. 1986).

biological coordination problem. Confirmation of critical fluctuations, critical slowing down and switching time predictions show that the emergence of coordinated movement patterns can be understood – quite precisely and consistently – in terms of the physics of nonequilibrium processes. We see also how the language and tools of synerget-

ics, when closely allied to experimental data, can offer a window into understanding pattern formation, including stability and change of pattern in a complex, biological system, In particular, we have demonstrated that changes in behavior are associated with loss of stability as observed by enhanced fluctuations and slowing down of the order parameter. But how general is this mechanism for effecting change? Obviously, biological systems are comprised of more than two states and more than two components. Thus there is a real need to move beyond the simple bistable, two-component situation that has been studied thus far (see also section 2). Nevertheless, we suspect that the key concepts of stability and loss of stability shown here to play a central role, are of quite general significance to understanding order and pattern formation in natural systems at multiple levels of organization. It may therefore be useful to summarize the insights gained from the foregoing experimental and theoretical research in a set of propositions that in turn may guide further study. The term 'behavioral pattern' below should be conceived quite broadly. The 'truth' of the following propositions must, of course, be individually evaluated in the various experimental systems observed and paradigms studied.

4. Theoretical propositions for behavioral pattern stability and change

(1) Behavioral patterns can be characterized by low dimensional collective variables (or order parameters) whose nature and dynamics are specific to biological functions and tasks.

(2) Experimentally well-defined behavioral patterns (reproducible, stationary over certain observation times) correspond to stable collective states (or attractors) of the order parameter dynamics. Behavioral patterns can change only from one collective state to another.

(3) Environmental context, task variables and certain biological boundary conditions act as parameters on the collective dynamics in the sense that they can modify the behavioral patterns but are themselves not dependent on these patterns. A parameter that moves the system through different collective states is a control parameter in the sense of synergetics. Such control parameters may be quite unspecific in nature, rather than detailed prescriptions for behavior.

(4) Fluctuations in the dynamics of the order parameters are conceptually important to reconcile the stability of behavioral patterns with

the ability to change behavioral patterns. Fluctuations establish time scales relations (see Kelso et al. 1987) that govern the switching dynamics among collective states, e.g., whether stationary or transient behavior is observed, whether multistability or averaging over multiple states is observed, and so forth.

(5) Loss of stability leads to behavioral change. Fluctuations allow the system to feel its stability and loss of stability by exploring the environment of a collective state. Fluctuations thus govern the process of behavioral change. How a system changes depends on the availability and relative stability of the system's collective states.

(6) The level of description used in experimental observations of behavioral patterns (e.g., neuronal, muscular or kinematic) is a question of convenience. Patterns at all levels are governed by the dynamics of collective variables. In this sense, no single level has ontological priority over any other.

(7) In a number of specific cases (which may nevertheless, be quite common) collective states (e.g., multiple phase and frequency-lockings) may arise from a coupling of self-excited nonlinear (limit cycle) oscillatory processes. The dynamics of such collective states are always low dimensional (cf. Proposition 1). In such cases, the relation between levels can be defined precisely, and the manner by which the degrees of freedom are compressed may be understood.

5. Perspectives and future directions

One can find many phase transition-like phenomena in the behaviors produced by biological systems (see Kelso and Schöner (1987) for a number of examples from the movement literature, both neural and behavioral). It may be more important, however, to use insights gained into organizational principles to motivate new problems i.e., to characterize them in a way that allows them to be attacked (and solved). A sample of some of the questions worthy of study in the future are the following:

– Are changes in behavior always associated with instabilities? Up to now, only spontaneous (so-called involuntary) changes in behavior have been shown to be governed by stochastic, nonlinear dynamics. Are environmentally induced or voluntary changes from one collective

state to another also governed by relative stability and fluctuations (e.g., easier in the direction of more stability than vice-versa)? If so, the same processes would be shown to play a major role in intentional switching (typically defined in non-dynamical language) as in spontaneous switching.

– Movement, or more generally, action is often intimately related to perception. For example, in the experiments reported in section 3, a metronome serves to impose a certain frequency on the movements. More precisely, then, it is the coupled environment–movement system from which an ordered state, or behavioral pattern is formed. The quite generally important question thus arises: what are the essential collective variables of this collective perception–action system? What are the dynamics of this relation? How can this relation be changed? In our work such perspectives have already arisen (Schöner and Kelso in press-a,b) but much remains to be done.

– In our experiments, we found a collective variable and its dynamics in synchronized, rhythmic movement. Obviously there are many functional movements that are not synchronized or not even rhythmic. Another task of generalization is thus to identify the variables and dynamics for these cases. Thus a study of entrainment and the relative stabilities of the different entrained states is crucial. For discrete movements, earlier work (e.g., Kelso et al. 1979) showed already that interlimb coordination is characterized by fixed temporal relationships (e.g., a trend toward simultaneity). Another example is trajectory formation, where invariants found in the geometry of the trajectories hint at the existence of fixed temporal relationships (relative phasing) among different joints of a limb (Soechting and Terzuolo 1986).

– Another direction of biologically relevant generalization poses new theoretical questions: How can one characterize patterns in multilimb movements? If many patterns coexist, what is their relation? Does a potential exist that governs the dynamics of several collective variables? Are there preferred transition paths from one pattern to another?

– Finally the rather detailed dynamical understanding of biological coordination in the language of synergetics, offers a unique opportunity to study such long-standing problems as learning, memory and adaptation. How does the stability of a collective state evolve during learning? Does learning a new state affect the stability of other states? Does the system increase its dimensionality during learning? Can one measure the different time scales involved as parameters defining the collective dynamics are themselves made dynamic?

6. Conclusions

A main aim of the present theoretical approach is to establish (and develop) synergetic concepts – originally formulated by H. Haken in a physical context–for complex biological systems. The extent to which synergetics must be tailored to the needs of biology and psychology is an open question. As we have emphasized, it is not a straightforward application (e.g., of the slaving principle), but rather requires a particular strategy that is outlined here (section 2). In the disciplines of biology and psychology, various phenomena such as multiplicity of patterns, perception–action coupling, behavioral change, learning and so on demand to be understood. In view of the fact that very intricate patterns (including deterministic chaos) may be generated by low-dimensional dynamics, it seems not unreasonable to suggest that such behaviorally complex phenomena may be encompassed by the present strategy. It is important to emphasize that the meaning of 'understanding' depends on the language used. In the present perspective, understanding is sought within the abstract level of the essential (collective) variables and their dynamics, regardless of scale or material substrate. The approach, nevertheless must be operational, that is, it must formulate dynamical laws for observable variables only and it must make experimentally testable predictions. By showing that behavioral patterns on several levels of description can be understood using the tools of nonlinear dynamics, our hope is that a common language may emerge. Thus the language barrier that currently exists among scientists who observe complex systems at very different scales of analysis may be overcome. Synergetics' recognition that the cooperation among many subsystems may be governed by the same principles irrespective of the subsystems was a major step in this regard. A theory of biological coordination ultimately rests on a reductionism to a minimal set of laws governing dynamic patterns, and not the materialistic reductionism that currently predominates.

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KNOWLEDGE
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Joan N. VICKE
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Vickers, J.N., 1
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Introduction

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* This paper was derive
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Franks, Shirl Hoffman
Author's address: J.N
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0167-9457/88/$3.50 © 1